

MediKiosk System Architecture & Role Guide

Comprehensive System Breakdown • SIH Presentation & Technical Architecture Documentation

Core Clinical Principle: *'AI collects. AI structures. AI highlights. The doctor decides.'*

Built for real-world out-patient hospital intake: multilingual voice assistance, ABDM / FHIR R4 compliance, and human-in-the-loop safety.

1. Admin Portal (Hospital Administration)

Purpose: Central operational control for hospital IT and medical directors to monitor patient throughput, configure clinical departments, manage doctor/nurse credentials, and verify ABDM / FHIR regulatory compliance.

- **Dashboard (/admin/dashboard):** Live metrics tracking total patient flow, average triage & intake wait times, active emergency alerts, and department bottlenecks.
- **Patients Master Index (/admin/patients):** Central registry across all kiosk sessions supporting record verification, manual overrides, and historical lookups.
- **Doctor Management (/admin/doctors):** Roster management with strict RBAC. Only Administrators can **Add New Doctors** or **Remove Doctors**, track live queues, and review clinical latencies.
- **Departments (/admin/departments):** Configures clinical wings (Cardiology, General Medicine, Orthopedics) ensuring automated intake routing to correct specialty queues.
- **Analytics (/admin/analytics):** Operational charts detailing intake duration, triage accuracy heatmaps, high-frequency symptoms, and AI extraction accuracy.
- **Languages Hub (/admin/languages):** Configures regional Indian language packs (Hindi, Tamil, Telugu, Kannada, Bengali, English) for kiosk voice and display.
- **Integrations Hub (/admin/integrations):** Interoperability control plane. Monitors OCR & Risk Stratifier microservices. Features **'Test Simulated HIE Dispatch'** generating FHIR R4 Bundles for ABDM Health Information Exchange.
- **Security & Audit Logs (/admin/security, /admin/audit-logs):** Immutable, timestamped audit trails of every clinician record access, consent modification, and export event.

2. Doctor Workspace (Physician Portal)

Purpose: Saves critical consultation minutes by presenting unstructured patient complaints as verified, decision-ready clinical insights before the patient enters the examination room.

- **Prioritized Queue (/doctor/queue):** Live patient queue dynamically ordered by Triage Risk Priority (Red Flag urgent cases automatically prioritized to the top).
- **Patient Story:** Cohesive, AI-generated clinical narrative synthesizing patient speech, answers, and complaint progression into medical SOAP notes.
- **Medical Timeline:** Visual chronological view of past hospital visits, chronic illnesses, and previous surgeries.
- **Extracted Structured Facts:** Discrete, verifiable tags for Symptoms, Onset/Duration, Current Medications, Known Allergies, and Baseline Vitals.
- **Risk Stratification & Red Flags:** Prominently flags urgent contraindications or clinical hazards (e.g., chest pain with hypertension, anticoagulant use with trauma).
- **Active Consultation:** In-encounter console to validate AI-structured findings, add physician notes, generate digital e-prescriptions, and discharge or refer.

3. Nurse / Triage Console

Purpose: Essential human-in-the-loop safety net bridging the automated patient kiosk with the physician consultation queue.

- **Triage Queue (/nurse/queue):** Staging area for patients who completed kiosk intake awaiting physical vitals verification.
- **Vitals Verification & Entry (/nurse/triage):** Nurse measures and records Blood Pressure, Pulse, SpO2, and Temperature, instantly linking with the AI intake record.
- **Urgent Red Flag Alerts (/nurse/alerts):** Real-time high-priority alerts triggered if kiosk dialogue detects acute emergency symptoms, enabling immediate physical intervention.

4. Patient Self-Service Kiosk

Purpose: Multilingual, high-accessibility self-service intake terminal designed for hospital waiting areas, replacing slow paper registrations with an empathetic AI conversation.

- **Welcome Screen (/patient/welcome):** High-contrast welcome portal with prominent **Easy Mode (Elderly Assist)** toggle.
- **Intense Easy Mode:** +55% enlarged typography, ultra-high contrast dark borders, clean layout wrapping, and full automated Text-to-Speech (TTS) guidance.
- **Language & Accessibility:** Voice guidance toggle and multi-lingual regional language support.
- **ABDM Consent Gate (/patient/consent):** Transparent, legally grounded digital consent capture complying with Indian Digital Personal Data Protection (DPDP) and ABDM standards.
- **Patient Identity (/patient/identity):** Flexible patient lookup via Mobile OTP, Hospital MRN, or Government-issued **ABHA Number**.
- **Conversational AI Intake (/patient/intake):** Dynamic clinical dialogue adaptively probing chief complaints with follow-up questions tailored to reported symptoms.
- **Document Intelligence Scanner (/patient/document-processing):** Optical Character Recognition (OCR) pipeline scanning physical paper prescriptions and lab reports.
- **Review & Token Handoff (/patient/review, /patient/complete):** Plain-language summary confirmation followed by clear queue token routing (e.g., 'Proceed to Nurse Station B').

5. Recommended SIH Presentation Demo Flow

To create maximum impact with the evaluation jury, demonstrate the system in the following 4-step sequence:

Step	Role / Persona	Key Demo Highlights
1	Patient Kiosk	Show Easy Mode toggle (+55% text & voice), log in with ABHA number, conduct quick voice intake, and scan a paper prescription.
2	Nurse Triage	Show incoming kiosk record, demonstrate how high-risk complaints trigger instant triage alerts, and input verified vitals.
3	Doctor Workspace	Highlight time savings: demonstrate AI Patient Story, structured facts, red-flag detection, and the core tenet: 'AI suggests, Doctor decides'.
4	Admin Portal	Show operational analytics, doctor roster management, and trigger the Test Simulated HIE Dispatch to prove live ABDM FHIR R4 interoperability.